

Basic Office Network Documentation

Week 2 GRC Portfolio Project

Company

ABC Solutions Pvt. Ltd.

Prepared By

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Version

1.0

Date

11 August 2026

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Purpose

This document provides a basic overview of the network infrastructure of ABC Solutions Pvt. Ltd. It documents the main network components, their functions, communication protocols, security controls, and common network security risks. The purpose is to provide a clear reference for understanding how the company's network is structured and how basic security measures are applied.

Scope

This document covers the small office network of ABC Solutions Pvt. Ltd., consisting of approximately 25 employees across HR, Finance, IT, and Operations departments. It covers the network devices, IP addressing, commonly used protocols, basic security controls, network security risks, and recommended security measures within the office network.

Company Network Overview

ABC Solutions Pvt. Ltd. is a fictional company with approximately 25 employees working across four departments: HR, Finance, IT, and Operations. The company operates from a small office and uses a local area network (LAN) to connect employee computers, laptops, printers, and other network devices. The office network provides employees with access to internal resources and the Internet.

Network Environment

The office network uses a router to connect the internal network to the Internet, a firewall to control network traffic, and a network switch to connect wired devices. A wireless access point provides Wi-Fi connectivity for laptops and other wireless devices. Employee computers and a network printer are connected to the internal network.

Information	Details
Company	ABC Solutions Pvt. Ltd.
Employees	Approximately 25
Departments	HR, Finance, IT, Operations
Network Type	Local Area Network (LAN)
Internet Access	Broadband Internet

Network Diagram

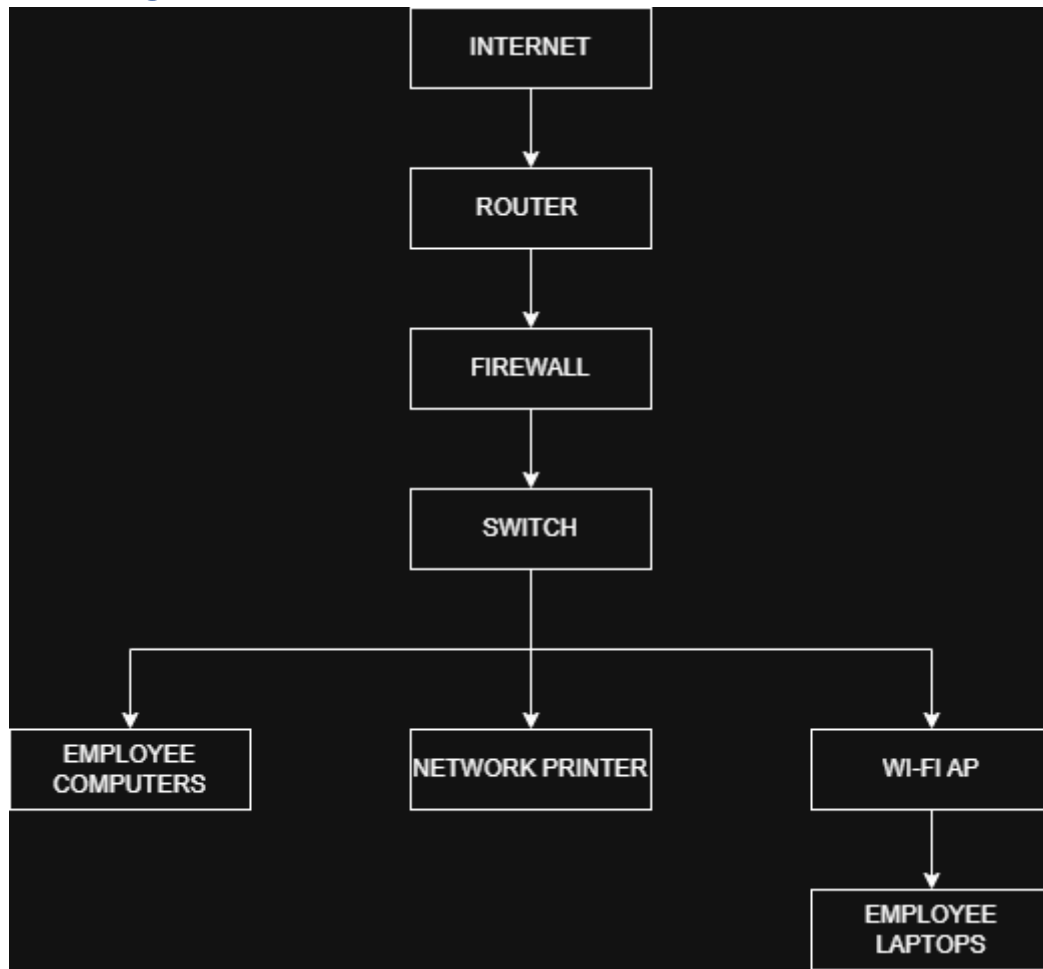


Figure 1: Basic Office Network Architecture

Network Components

Component	Purpose	Security Relevance
Internet	Provides external network connectivity for the office.	Provides access to external services and can expose the network to external threats.
Router	Connects the office network to the Internet and routes network traffic.	Should be securely configured and protected with strong administrator credentials.
Firewall	Monitors and controls network traffic based on security rules.	Helps block unauthorised network connections and unwanted traffic.
Network Switch	Connects devices within the local office network.	A managed switch can support access controls and network segmentation.
Employee Computers	Provide employees with access to business applications and resources.	Should use antivirus, security updates, and access controls.
Network Printer	Allows employees to print documents over the network.	Should be protected from unauthorised access and kept updated.

Wi-Fi AP	Provides wireless network connectivity to laptops and other devices.	Should use strong Wi-Fi security and authentication.
Employee Laptops	Allow employees to access company resources through the office network.	Should use strong authentication, security software, and regular updates.

These components work together to provide network connectivity while supporting basic security controls. The router and firewall manage communication with external networks, while the switch, wireless access point, computers, laptops, and printer support internal office connectivity.

IP Addressing

The office network uses private IPv4 addresses for communication between internal devices. The following addresses are examples created for this documentation and do not represent an actual company network.

Device	Example IP Address	Purpose	Network Role
Router	192.168.1.1	Connects the internal network to the Internet.	Gateway
Firewall	192.168.1.2	Filters and controls network traffic.	Security Control
Network Switch	192.168.1.3	Connects devices within the local network.	LAN Connectivity
Wi-Fi AP	192.168.1.4	Provides wireless network connectivity.	Wireless Access
Employee PC-01	192.168.1.101	Employee workstation.	End Device
Employee PC-02	192.168.1.102	Employee workstation.	End Device
Network Printer	192.168.1.150	Provides network printing services.	Peripheral Device
Employee Laptop-01	192.168.1.151	Provides an employee with wireless access to company resources.	End Device

Note: The IP addresses shown above are private IPv4 addresses used only for this fictional network documentation. Actual IP addressing may vary depending on the organisation's network design and configuration.

Network Protocols

Network protocols define how devices communicate and exchange information over a network. The following protocols are commonly used to support web access, name resolution, IP address assignment, secure administration, and email communication.

Protocol	Purpose	Example	Security Consideration
HTTP	Transfers web content between a client and a web server.	Visiting a website using HTTP.	Data is not encrypted, so HTTPS is preferred for sensitive communication.
HTTPS	Provides secure web communication using encryption.	Online banking or shopping.	Helps protect data from interception during transmission.

DNS	Converts domain names into IP addresses.	Converting <code>example.com</code> into an IP address.	DNS security and trusted DNS servers help reduce malicious redirection risks.
DHCP	Automatically assigns IP addresses and network settings to devices.	A laptop receives an IP address when joining office Wi-Fi.	Unauthorised DHCP servers can provide incorrect network settings.
SSH	Provides secure remote access to network devices and servers.	An administrator remotely connecting to a server.	Should use strong authentication and restricted access.
FTP	Transfers files between computers over a network.	Uploading files to an FTP server.	Standard FTP does not encrypt credentials or data.
SMTP	Sends email messages between mail servers and email clients.	Sending a business email.	Secure configurations and authentication help protect email communication.

Network Security Controls

Network security controls are measures used to protect the office network, systems, and data from unauthorised access, attacks, and other security threats. The following controls provide basic protection for the fictional office environment.

Security Control	Purpose	Location	Security Benefit
Firewall	Filters and controls incoming and outgoing network traffic.	Between the Internet and internal network	Helps prevent unauthorised network connections.
Antivirus	Detects and helps remove malicious software.	Employee computers and laptops	Reduces the risk of malware infections.
Multi-Factor Authentication (MFA)	Requires more than one method to verify a user's identity.	User accounts	Reduces the risk of account compromise from stolen passwords.
VPN	Provides a secure connection to the organisation's network over the Internet.	Remote employee connections	Helps protect remote access and data in transit.
Network Segmentation	Separates network resources into different logical sections.	Internal network	Helps limit access and reduce the impact of a security incident.
Access Control	Restricts access to systems and resources based on user permissions.	Systems and network resources	Prevents unauthorised users from accessing sensitive resources.

Security controls should be selected and implemented based on the organisation's risks, business requirements, and security needs. Controls should also be regularly reviewed to ensure they remain effective.

Network Security Risks

Network security risks are potential events that could negatively affect the confidentiality, integrity, or availability of the organisation's systems and information. The following risks are based on the fictional office network described in this document.

Risk	Threat	Vulnerability	Possible Impact	Recommended Control
Unauthorized Network Access	External attacker	Weak network access controls	Unauthorized access to company resources	Firewall and access controls
Malware Infection	Malware	Unpatched or unprotected employee devices	Data loss or system disruption	Antivirus and regular security updates
Data Interception	Attacker	Unencrypted communication	Exposure of sensitive information	Use HTTPS, SSH, and VPN
Wi-Fi Unauthorized Access	Unauthorised user	Weak Wi-Fi security	Exposure of sensitive information	Strong Wi-Fi authentication and encryption
Denial-of-Service Attack	Attacker	Exposed network services	Network or service unavailability	Firewall rules and network monitoring

The identified risks should be assessed according to their likelihood and potential impact. Appropriate security controls can then be implemented to reduce the organisation's exposure to these risks.

Security Recommendations

The following recommendations are intended to reduce the identified network security risks and strengthen the overall security of the office network. These measures should be implemented according to the organisation's business requirements and risk level.

No.	Recommendation	Related Risk
1	Use strong passwords and Multi-Factor Authentication (MFA) for user accounts.	Unauthorized Network Access
2	Keep operating systems, applications, and network devices updated with security patches.	Malware Infection

3	Use HTTPS, SSH, and VPN for secure communication where appropriate.	Data Interception
4	Secure the office Wi-Fi using strong authentication and encryption.	Wi-Fi Unauthorized Access
5	Configure firewall rules and monitor network activity for suspicious traffic.	Denial-of-Service Attack

Implementing these recommendations can help ABC Solutions reduce common network security risks and improve the protection of its systems, users, and information. Security measures should be reviewed periodically as the organisation's network and business requirements change.

Version History

Version	Date	Author	Changes
1.0	11 August 2026	Durva Amin	Initial Release